UC BERKELEY SOLAR VEHICLE TEAM

Dynamics - Gen XI Pedal Box

Critical Design Review

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Constraints and Goals

- Enhance wall mounting system
  - Improve stability, ease of installation, and compatibility with the new chassis.
- Reduce overall weight without compromising durability
  - Optimize material selection and geometry, minimizing mass while maintaining or exceeding Excalibur durability standards.
- Above min FOS of 1 when applied forces (1000N) are normal
- Determine forces going into occupant cell walls

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Manufacturing Plan

- Carbon fiber layups
  - create panel to be sandwiched by mounting systems of gas and brake pedals
  - Support panel with 45 degree angle CF
- Pedal pads
  - Reusing Excalibur's design → CNC 6061-t6
- Mounting system and pedal arms
  - Water jet with 6061-t6 aluminum
- Screws and springs
  - bought through McMaster (max load of spring = 73.26 lbs)
- Inserts
  - CNC'd 6061-t6 aluminum
- After screwing together the mounting system and pedals and sandwiching the carbon fiber panels → entire piece can be epoxied anywhere

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Manufacturing

Part Name	Quantity	Manufacturer
Gas Pedal Arm	1	Waterjet+Mill - Student
Gas Pedal Pin	1	Purchase + Mill - Student
Gas Side Panels	2	Waterjet - Student
Torsion Spring	4	McMaster
Gas Carbon Fiber Panel	1	Layups - Student
Crossbar	1	Lathe - Student
Inserts	7	Lathe - Student
Brake Side Panel	2	Waterjet- Student
Brake Pedal Arm	1	Waterjet - Student
Brake Pedal Pin	1	Purchase+Mill
Brakes CF Panel	1	Layup - Student
Brake Pedal Pad	1	Tormak - Student
Gas Pedal Pad	1	Tormak - Student

Bill of Materials

Part Name	Manufacturer	Quantity	Price
Grip Tape	McMaster-Carr	1	$10.49
Stainless Steel Pan Head Phillips Screw(10-32 Thread Size, 11/16" Long)	McMaster-Carr	1	$13.69 per pack of 100
SENSOR ROTARY 90DEG CONNECTOR	Digikey	2	$76.66*2=$153.32
600-Series Balance Bar Assembly	Tilton	1	$74
0.05 in thick 1/2 in dia washers	McMaster-Carr	1	$9.74
90 degree Left-hand wound 0.989" OD torsion spring	McMaster-Carr	4	$7.59*4=$30.36
Pedal Box Bolts-Passivated 18-8 Stainless Steel Pan Head Phillips Screw	McMaster-Carr	1	$13.69
Black-Oxide Alloy Steel Socket Head Screw(stopper)	McMaster-Carr	1	$13.73
Zinc-Plated Medium-Strength Steel Hex Nut(stopper)	McMaster-Carr	1	$9.58
Short-Thread Alloy Steel Shoulder Screw	McMaster-Carr	6	$6.34*6=$38.04

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Full Assembly

Index	Part Name
1	Brake Foot Pad
2	Brakes Pedal Arm
3	Tilton 600 Series Balance Bar (7/16" Bar Diameter, 5/16" Master Cylinder Pushrod Thread)
4	Tilton Master Cylinder (Bore size ⅝")
5	Brakes Brackets
6	Carbon Fiber Panel
7	6061-T6 Aluminum Crossbar
8	Accelerator Brackets
9	McMaster Carr Black-Oxide Alloy Steel Socket Head Screw (5/16"-18 Thread Size, 1-1/2" Long)
10	Accelerator Pedal Arm
11	Accelerator Foot Pad
12	McMaster Carr Steel Extension Spring with Loop Ends (3" Long, 0.75" OD, 0.105" Wire Diameter)
13	Aluminum Pivot Pins
14	Honeywell Rotary Position Sensor

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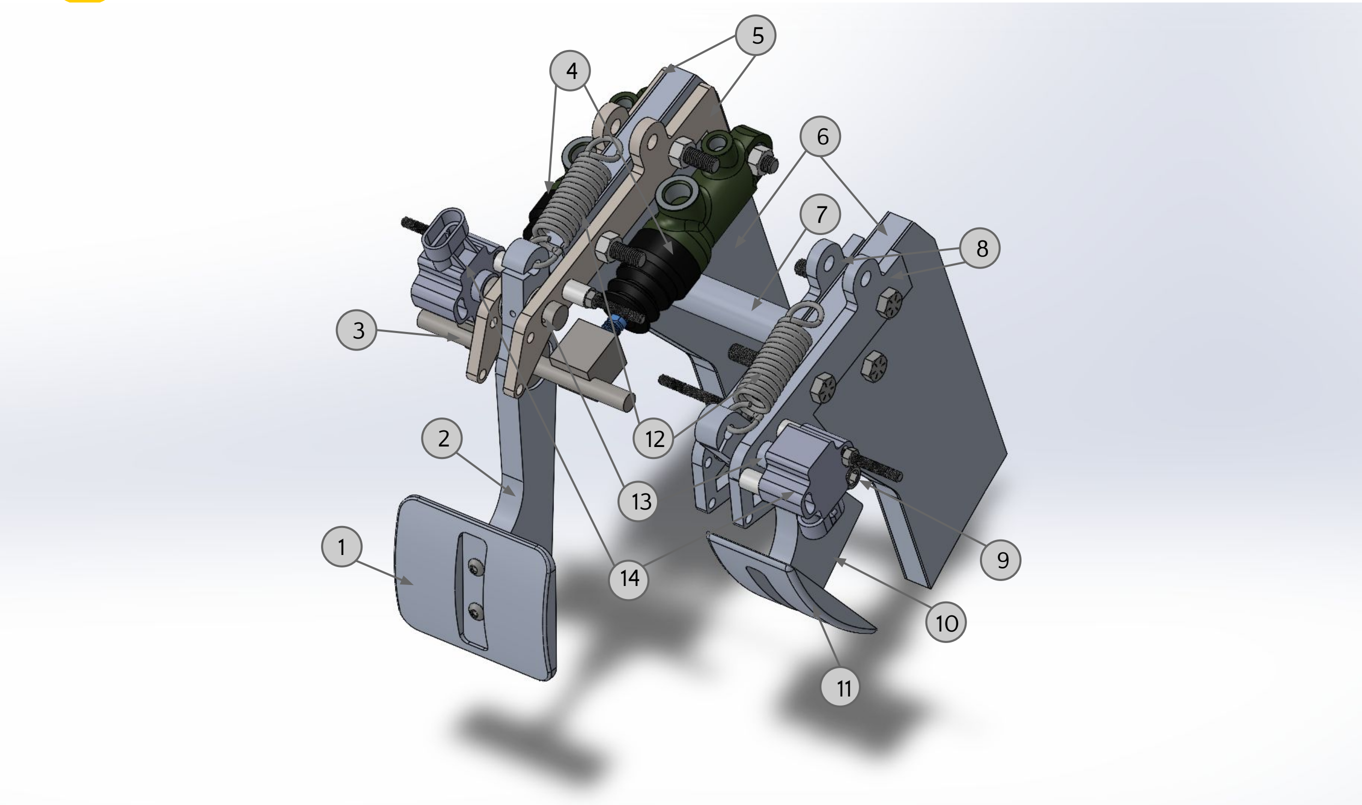
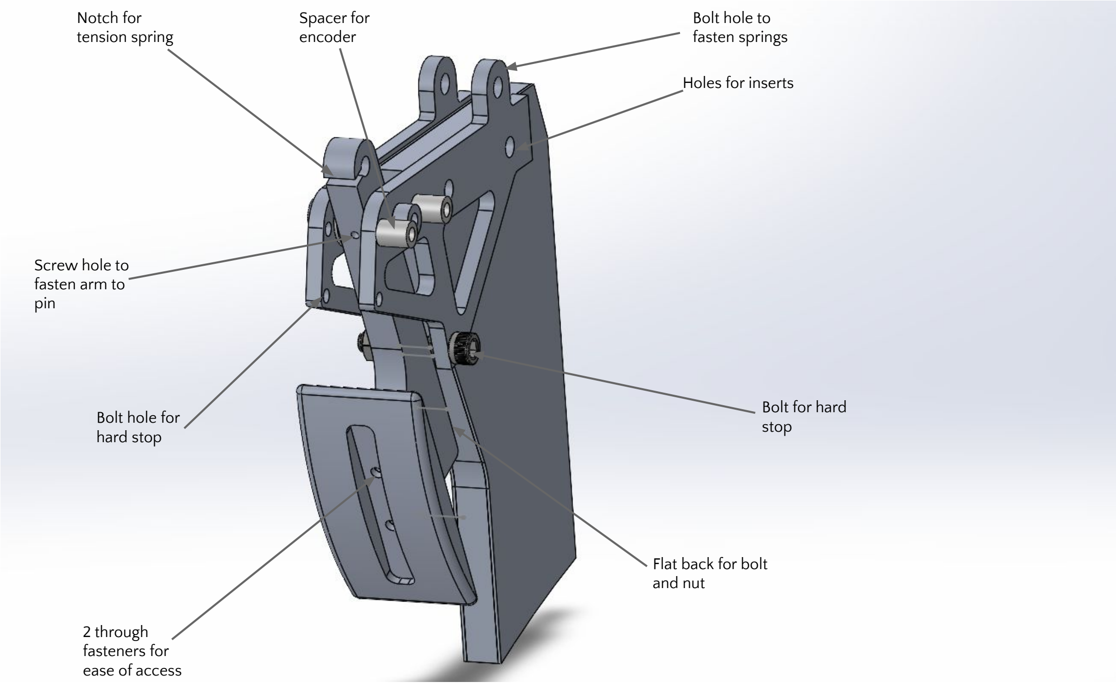
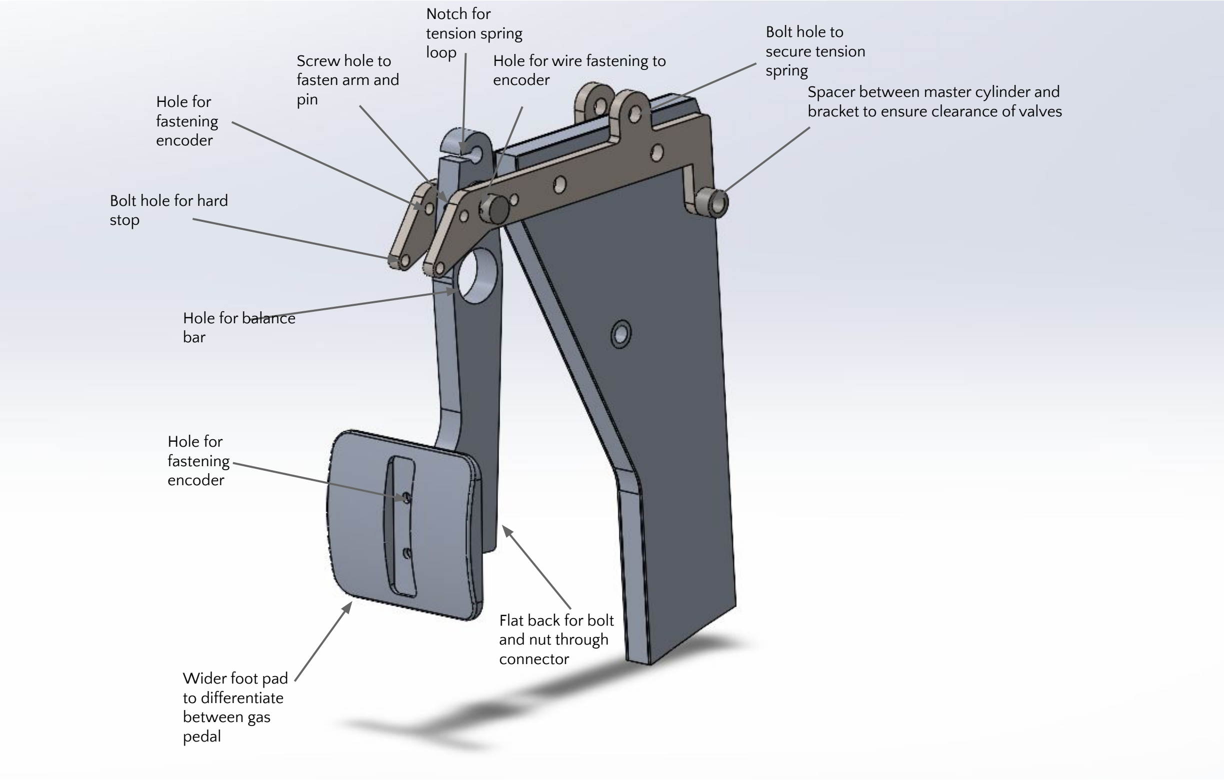


Design Decisions (Gas)



Design Decision (Brakes)



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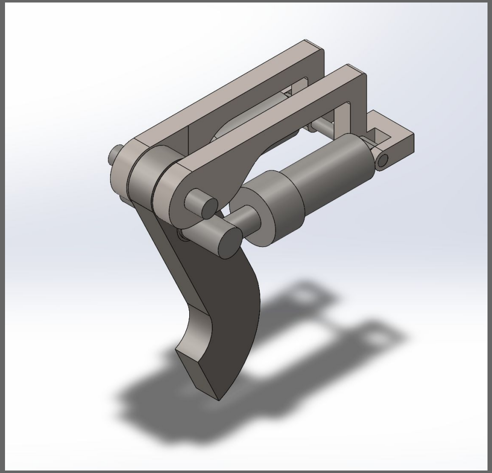
Modifications

- Decided to purchase:
  - Balance bar
- Make all bracket parts flat for ease of water jet manufacturing
- CF panel shape:
  - Now mounting to front bulkhead so back side of panel is now a right angle
  - Shorten height of gas assembly’s panel to save material/weight and make space for encoders
- Reduction of gas pedal bracket
  - brake pedal bracket height restricted b pedal ratio



Previous Version

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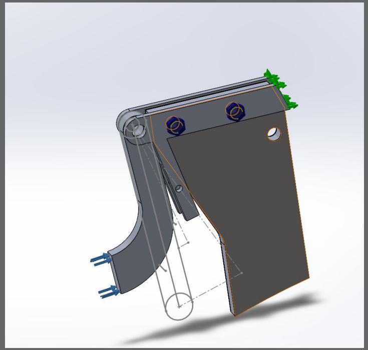
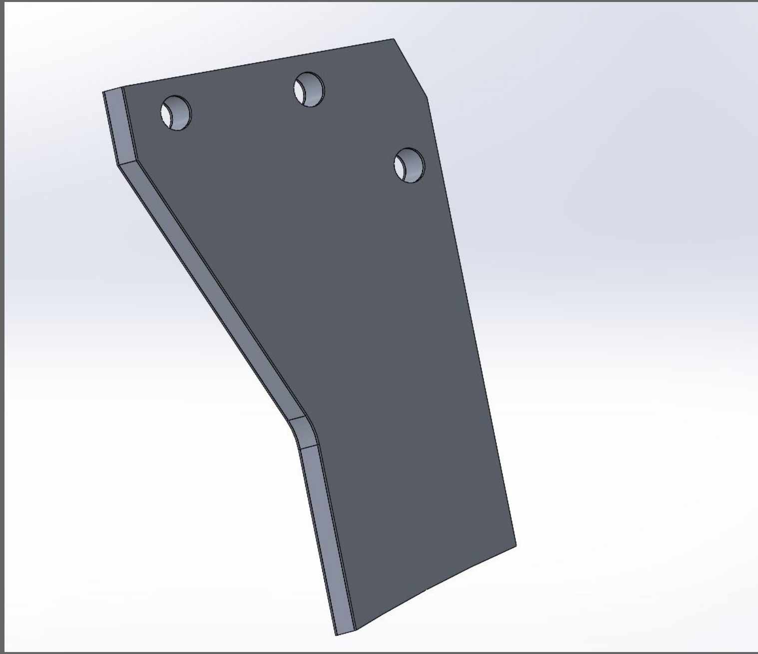
Simulation Modifications

- Previously did part by part simulations
  - resulting in abnormally high forces → meshed improperly due to parts contacting each other
  - These sims called for full steel construction
- New simulations
  - Separated into pedal arm, pin, and side panel assembly sims
  - Much more reasonable resultant forces, calls for aluminum



Carbon Fiber Panels

- 0.6” thick carbon fiber panel between the two side panels of both gas and brake systems
- 1/2” Aluminum honeycomb core
- Use of shoulder bolts to prevent peeling
- Because panel can be glued anywhere on occupant cell wall/floor with epoxy and the rest of the mounting system is bolted to the panel, pedal box components is very flexible in terms of location

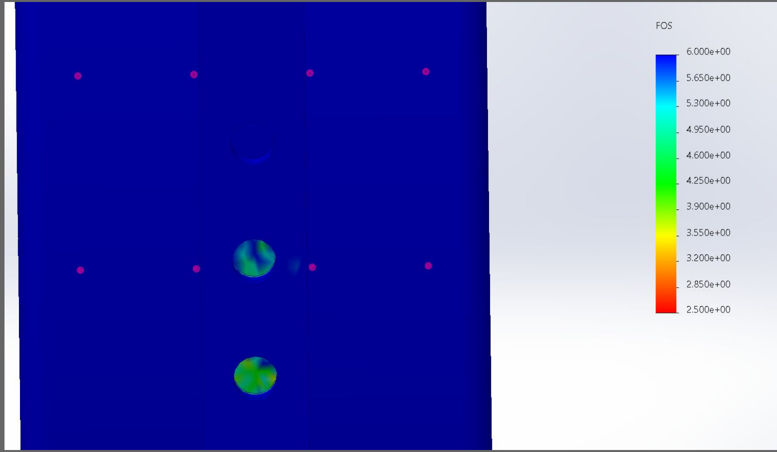


Technical Notes: Brakes

- Pedal Ratio: 4.35
  - 1.1 in travel distance for master cylinder means 4.785 in travel for the driver’s foot for full actuation
- Master cylinder bore size: 5/8in

Gas: Bolt Calculations

- Shoulder diameter: 0.19“
- Head diameter: 0.373”
- Thread size: 24 threads per inch
- Tensile strength: 70,000 psi
- Material: AISI304
- Max Torque: 16.69 Nm
- Preload in sims: 2Nm



Parameter	Value	Units
Bolt Material # from Material Table	86	
Yield Strength, Sty	896.32	MPa
Thread Size	5/16"-18	-
Nominal Diameter, dnom	6.35	mm
Tensile Stress Area, At	0.2347	cm^2
Preload % of Yield, %yld	50	%
Torque Coefficient, KT	0.2	-
Preload Uncertainty, %uncrt	25	%
Preload Relaxation, %relax	10	%

Bolt Selection

Inch ## Metric ##

Results

Preload Force, FPL (N)	Install Torque, T (N·m)	% of Yield Strength	Torque Coefficient, KT
10,517	13.3565	50	0.2

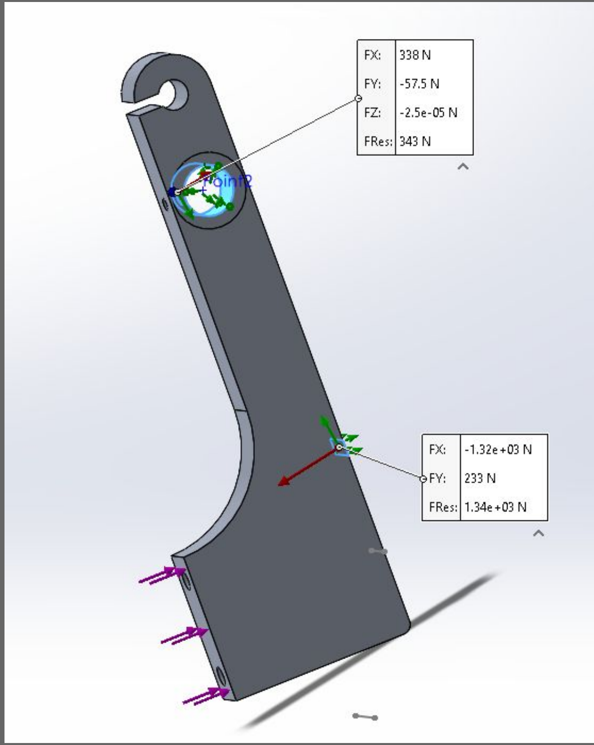
The nominal value is the design target, and the min and max values account for preload uncertainty and relaxation.

	Nominal	Minimum	Maximum
Preload Force (N):	10,517	6,836	13,146
% of Yield Strength	50	32.5	62.5
Install Torque, T (N·m)	13.3565	8.6817	16.6956

Brake: Bolt Calculations

- Bolt pretension: 5Nm
- Tensile strength: 105,000 psi
- Material: alloy steel
- FOS:1.3

Gas Pedal Arm FEA



Applied force: 1000 N

FOS 6.2

Fixed geometry on pedal pin, roller on stopper block

Coords: x = front of car, y = up, z = right of car

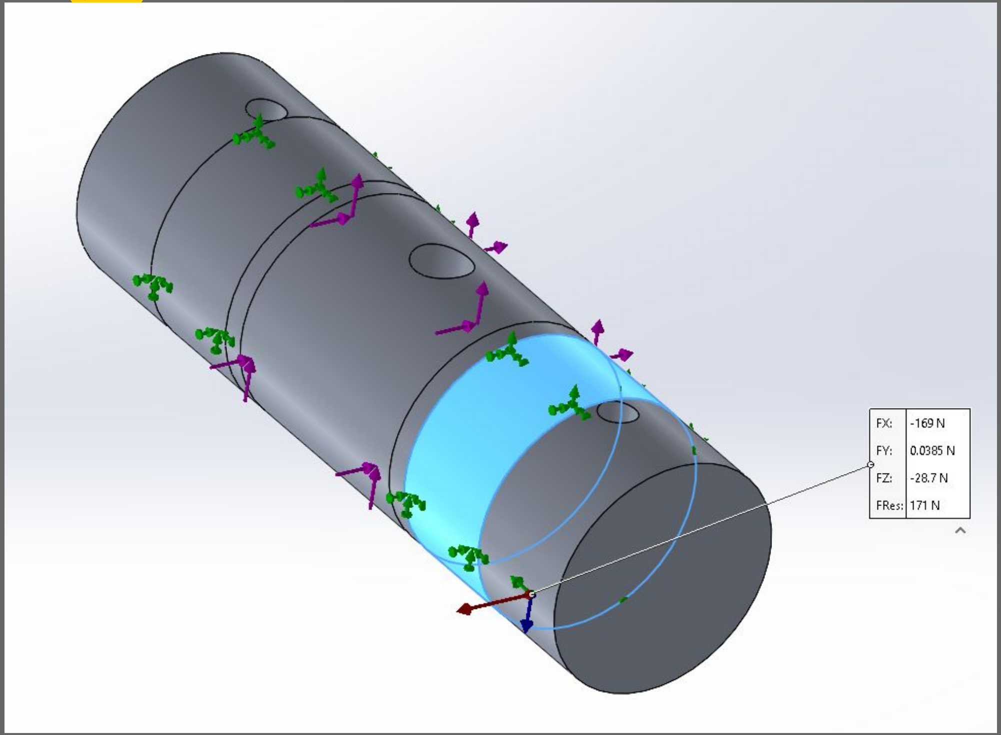
Resultant Forces on pin:

338N, 57.5N

Resultant Forces on Stopper:

-1.32x10ˆ3 N, 233N

Gas Pedal Pin FEA



Applied forces: 338Nx, 57.5Ny

FOS 54

Fixed geometry on side panel regions, applied force on pedal arm region

Coords: x = front of car, y = left of car, z = up

Resultant Forces on side panel:

-169N, 0.0385N, -28.7N

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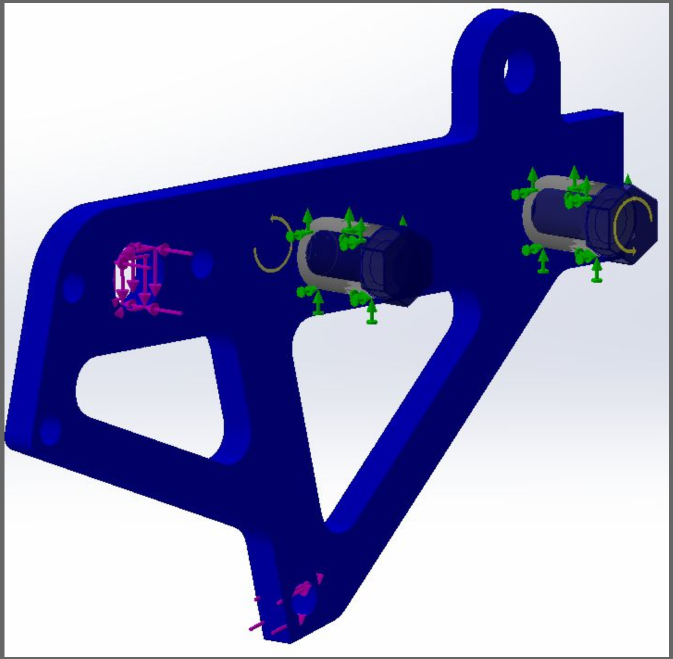
Gas Brackets FEA



Applied forces: -169Nx, 0.0385Ny, -28.7Nz

FOS 1.9

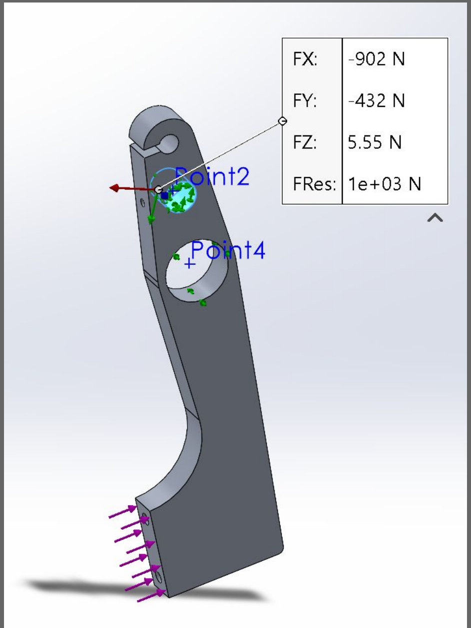
Fixed geometry on inserts

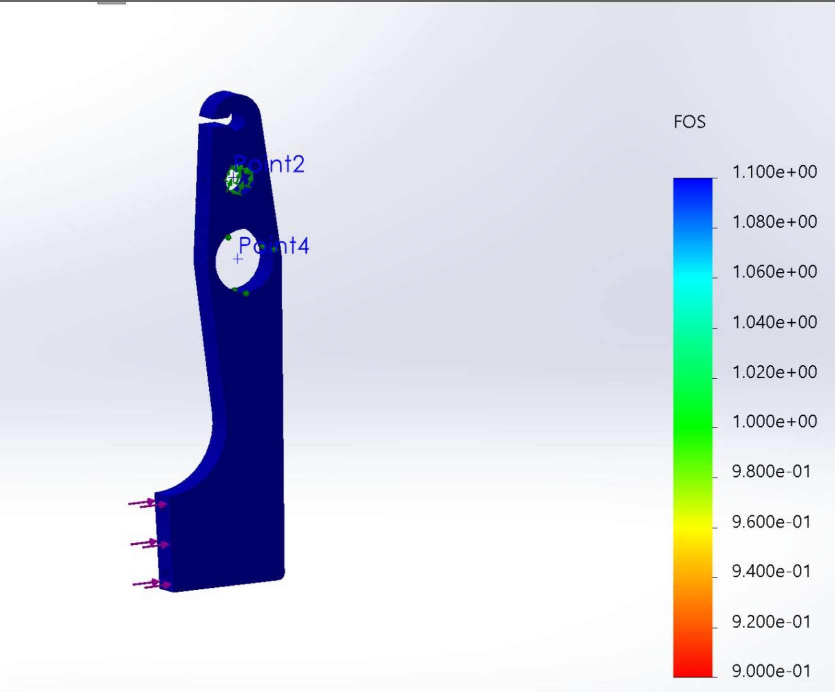
Coords: x = front of car, y = left of car, z = up

Bolt connections through inserts

- Connection interactions: contact

Brake Pedal Arm FEA





1000 N

FOS 1.3

Fixed geometry on pedal pin, master cylinder

Coords: x = front of car, y = up, z = right of car

Resultant Forces on pin:

-902N, -432N, 5.55N

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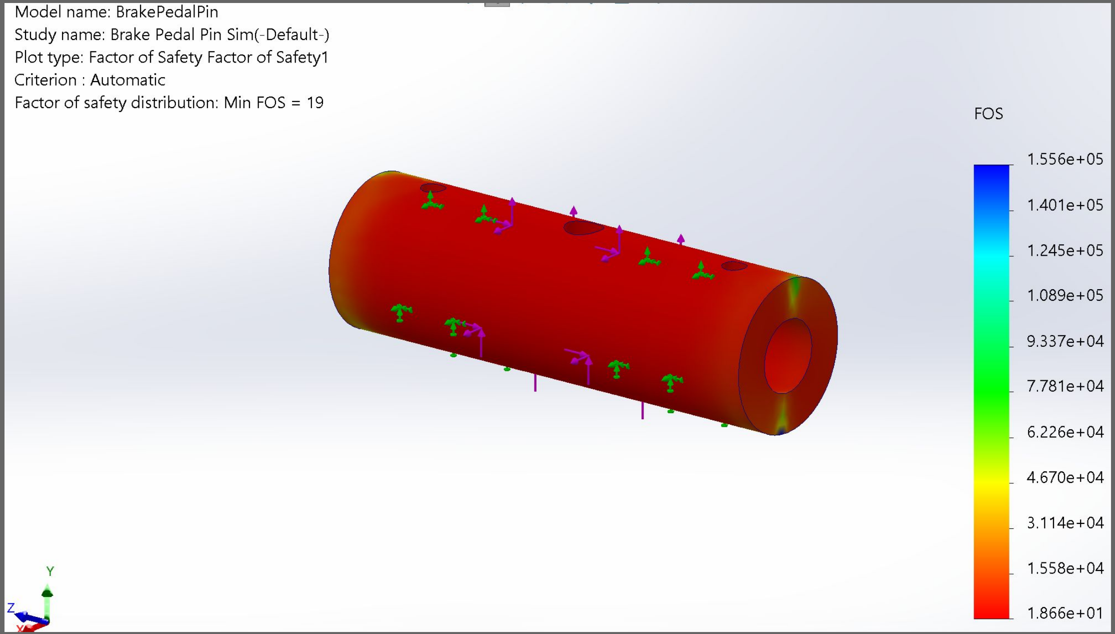
Brake Pedal Pin FEA



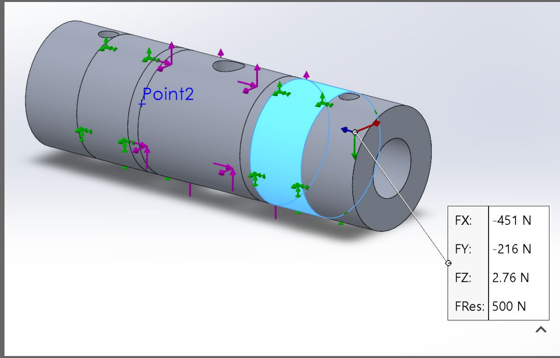
FOS 19

Fixed geometry on side panel regions, applied force on pedal arm region

Coords: x = back of car, y = up, z = left of car

Resultant Forces on side panel:

-451N, -216N, 2.76N



Potential Concerns and Mitigations

Breakage between inserts and carbon fiber panel → ensure proper sizing of inserts (not too small)

Ergonomic spacing between gas and brake pedal → 3D printed prototype on linear rails for driver feedback on comfort

Ensure proper return of pedal after use → tension springs bring pedal arm back to rest position

Constraining balance bar with a retaining ring to prevent lateral movement

Steering column going in between the brake and gas pedals → enough clearance for 1 foot braking but will consider driver feedback on prototype

Need for redundancy → cleared with electrical team

Fastening of encoder and encoder mounting → additional wire fastening included